stabilis — progress report

Frequency stability analysis engine. Milestones M0, M1 and M2 complete — the v0.1 line, the first version from which an uncertainty may legitimately be quoted — and most of M3: Greenhall EDOF, which is error-bar parity with Stable32; the exact non-negative least squares core and the power-law decomposition built on it; TOTVAR, the Thêo family and the MODWT wavelet variance. M5 is complete ahead of its place in the sequence, being the part with no equivalent in any other tool: the bench model, the cross-variance, and multi-channel noise rejection. M4's fitting layer has since landed on the same principle — the covariance *between* τ is measured rather than assumed away, so a fitted parameter's uncertainty is the one it actually has. M4's export layer has landed with it: the record back out as self-describing ASCII, the curve as CSV or JSON, the block Stable32 prints — reproduced character for character against its own committed exports — and a bundle directory whose digest covers the measurement rather than the files that carry it.

`generated 2026-08-15 23:54 UTC · stabilis 0.0.1.dev0 · numpy 2.2.6 · python 3.10.12`
`commit 2969c94 (61 total) — WORKING TREE DIRTY · every figure below is measured at generation time`

What this is

Stable32 is the de-facto tool for frequency stability analysis. It is Windows-only, closed, GUI-driven, unscriptable and no longer developed. *stabilis* is an open-core engine for the same job, intended as the free foundation under a paid workbench that produces signed ISO-17025-style reports for calibration labs and oscillator manufacturers.

Two things distinguish it from a reimplementation. First, the Haar wavelet variance *is* the Allan variance, so classical AVAR is the K=1 case of one framework; higher-order Daubechies filters then buy drift immunity and convergence for steeper noise. Second, confidence intervals are derived from the estimator's own coefficient autocovariance, so they do not require identifying the noise type first — Stable32 looks it up in a table and has no answer when the noise is not one of its five power laws.

> **The commercial argument for validation.** ISO/IEC 17025:2017 clause 6.4.1 defines software as *equipment*, and 7.11.2 requires laboratory software to be validated before introduction. Its NOTE 2 exempts commercial software "in general use within its designed application range" — an exemption stabilis fails twice over, being new and being open-core, since a lab that patches it explicitly triggers re-validation. An accredited lab therefore cannot adopt this engine unless it ships validation evidence. That evidence is the product's moat, not its overhead.

Results: accuracy against certified values

Every classical variance is a sum of squares of integer linear combinations of the phase samples, divided by an integer. Both NBS14 reference sets are integer-derived, so on them the variance is an **exact rational**. It is computed independently in `fractions.Fraction`, with the deviation certified to 50 digits via `decimal.Decimal.sqrt()`. That gives certified values computed here, to far more digits than any published table, with no dependence on a transcription being correct.

Accuracy is reported as **LRE**, the log relative error, after the NIST Statistical Reference Datasets convention: it is the number of correct significant digits. float64 carries about 15.95, so these are at the representation limit.

Estimator	NBS14 9-point (min LRE)	NBS14 1000-point (min LRE)
adev	16.3	14.8
oadev	16.1	14.8
mdev	16.3	14.1
tdev	16.0	14.1
ohdev	16.2	14.5

A worked certified value

Overlapping Hadamard variance of the NBS14 nine-point set at AF 1, as an exact rational and as a 50-digit deviation:

exact variance	`10027 / 2`
certified deviation	`70.806073185850378720054399356481405765799283518283`
computed (float64)	`70.80607318585038`

> **The oracle earned its keep immediately.** The relayed NBS14 table gave `hdev(1) = 70.80608` while listing `ohdev(1) = 70.80607` — but these are the same estimator at AF 1, since there is no overlap to exploit. The certified value settles it at `70.806073185850378`, so the first was a transcription error. This is exactly why correctness rests on algebra rather than on published tables.

Results: analytic identities

These need no reference data and no oracle, which is why they carry the correctness argument. Every published ADEV table traces back to Stable32, so agreeing with one proves interoperability; agreeing with algebra proves correctness.

Identity	Claim	Result
linear frequency drift: avar = $D^2 \cdot \tau^2/2$	bit-exact	**PASS**
linear frequency drift: hvar = 0	bit-exact	**PASS**
constant frequency offset leaves avar unchanged	bit-exact	**PASS**
scale equivariance at 2^k: var → 64·var for c = 8	bit-exact	**PASS**
adev = oadev = mdev at AF 1	rel 1e-15	**PASS**
tdev = $\tau/\sqrt{3}$ · mdev	rel 1e-15	**PASS**
constant phase gives exactly zero	bit-exact	**PASS**

> **The most valuable single test.** For a pure linear frequency drift D, the phase is quadratic, so the second difference is the constant $D \cdot \tau^2$ and the third difference vanishes. Hence `avar = D²τ²/2` and `hvar = 0`, both exactly. One test validates the Allan normalization, the Hadamard normalization and the drift convention simultaneously — and it is bit-exact, not approximate.

Results: interoperability with Stable32

Labelled as interop, not correctness: these published values were produced by Stable32 itself. Seven significant figures as published, so agreement is assessed at rtol 1e-6.

Estimator	AF	computed	published	rel. diff
adev	1	2.9223188e-01	2.9223190e-01	7.5e-08
adev	10	9.9657361e-02	9.9657360e-02	6.3e-09
adev	100	3.8978043e-02	3.8978040e-02	8.5e-08
oadev	1	2.9223188e-01	2.9223190e-01	7.5e-08
oadev	10	9.1599534e-02	9.1599530e-02	4.6e-08
oadev	100	3.2413430e-02	3.2413430e-02	8.0e-09
mdev	1	2.9223188e-01	2.9223190e-01	7.5e-08
mdev	10	6.1723764e-02	6.1723760e-02	6.2e-08
mdev	100	2.1709209e-02	2.1709210e-02	4.0e-08
tdev	1	1.6872015e-01	1.6872020e-01	2.8e-07
tdev	10	3.5636232e-01	3.5636230e-01	4.7e-08
tdev	100	1.2533818e+00	1.2533820e+00	1.8e-07
ohdev	1	2.9438833e-01	2.9438830e-01	9.9e-08
ohdev	10	9.5810832e-02	9.5810830e-02	1.8e-08
ohdev	100	3.2376383e-02	3.2376380e-02	7.8e-08

Results: reference generator and numerics

NBS14 1000-point set ships as arithmetic, not a file

The set is defined by a published Lehmer recursion `k[i+1] = 16807·k[i] mod 2147483647` from seed `1234567890`, so it needs no data file and carries no licence question. Verified against the published `wriley.com/tst_suit.dat`: all 1000 values agreed to 5.55e-16 — one ULP, which is that file's own text rounding. **The generator is the more exact artifact of the two.**

k[0]	`1234567890`
k[1] (check value)	`395529916` PASS
vs published digits	`max dev 3.33e-16`

Phase ↔ frequency round-trip

Recovers the offset-removed phase to a few ULP. Not bit-exact: a cumulative sum of differences does not telescope exactly in floating point. The test bounds this at 8 ULP, a figure derived from the measurements below rather than tuned.

n	peak (s)	max error (s)	ULP
10	2.690e-12	1.010e-28	0.2
1000	5.009e-11	6.462e-27	1.0
100000	2.415e-10	1.292e-25	2.5

Power-law noise synthesis

Kasdin–Walter generation, log-log periodogram slope recovered over 1e-3 < f < 0.1, averaged over eight realizations.

α (requested)	slope (recovered)	|error|
2	+1.9734	0.0266
1	+0.9912	0.0088
0	+0.0043	0.0043
-1	-0.9943	0.0057
-2	-1.9955	0.0045

Results: one missing sample

Gap handling is where a metrology tool fails most expensively, because it fails quietly. Integrating frequency to phase with a cumulative sum cannot step over an unknown increment, so a single `NaN` made every later phase sample `NaN` — and the estimator, dropping non-finite terms as designed, silently analysed only the data *before* the gap.

Measured at generation time over 400 realizations of 200 white-FM points with one sample removed at the midpoint, each compared against its own ungapped answer. *Old* integrates the gapped record in one cumulative sum, which is what the estimator used to do; *new* segments at the gap, integrates each run from its own zero and pools the coefficients. Bias is the mean signed error and is the figure that matters — on any single record either path can land closer by luck, but only one of them is centred on the right answer.

m	old bias	new bias	old RMS	new RMS	terms ungapped / old / new	terms lost
1	-0.04%	+0.01%	5.92%	0.57%	199 / 99 / 197	2 = 2m
2	-0.50%	+0.03%	6.78%	0.77%	197 / 97 / 193	4 = 2m
4	-0.86%	-0.10%	8.82%	1.52%	193 / 93 / 185	8 = 2m
8	-1.25%	-0.18%	12.25%	3.27%	185 / 85 / 169	16 = 2m
16	-3.00%	-0.29%	18.16%	6.92%	169 / 69 / 137	32 = 2m

> **One sample in two hundred, and the old path was biased low by percent — with no warning and a plausible-looking term count.** The bias is systematic and grows with τ, because the surviving terms were not a random subset of the record: they were the first half of it. Scatter falls by roughly an order of magnitude too, which is the direct consequence of keeping 197 terms instead of 99.
> The correct cost of a gap is the `2m` terms whose window spans it — `3m` for the Hadamard and modified families — and that is now exactly what it costs, asserted term by term in `tests/test_gaps.py`. Frequency input defaults to the segmenting policy and refuses the masking one; phase input keeps masking, which is correct when the timescale ran continuously across the gap. The choice, the number of runs and the longest run are all recorded in provenance.
>
> The residual few tenths of a percent in the "new bias" column is not a defect and does not go away: `E[σ̂] = σ·√(2/η)·Γ((η+1)/2)/Γ(η/2)`, so a deviation estimated from fewer terms is genuinely low on average. It is the same bias the ungapped estimate carries, slightly amplified. Correcting it, and reporting the error bar that goes with it, is what M2 went on to do — and the reason this defect had to be fixed first, since it feeds the term count that sets η.

Results: seeing under the instrument

Every estimator here is a mean of squared coefficients. Measure one clock through two independent channels and replace one factor with the other channel's coefficients, and the instrument noise stops being a floor: writing $x_a = x_c + n_a$ and $x_b = x_c + n_b$, the auto-variance of either is $\sigma^2_c + \sigma^2_n$ however long you log, while $E[d_a d_b] = E[d_c^2]$ exactly, because the two cross terms have expectation zero. Neither Stable32 nor `allantools` computes this.

Measured at generation time over 120 realizations of a white-FM clock seen through two white-PM channels, 8192 points each. Both columns are ratios to *the clock record's own* $\sigma^2(\tau)$ — not a closed form and not a fit — so the table is the rejection itself rather than a consistency check. A ratio of 1.00 is the right answer; the auto column is what a single channel would have told you.

τ (s)	auto / truth	cross / truth	ensemble spread	reported ±	rejection
1	8.64×	1.014×	12.1%	13.1%	18.2 dB
2	4.81×	1.015×	6.9%	7.4%	18.1 dB
4	2.91×	1.004×	4.0%	4.8%	17.8 dB
8	1.96×	1.004×	2.4%	4.4%	16.5 dB
16	1.48×	1.001×	1.6%	5.4%	14.4 dB
32	1.24×	1.001×	1.1%	7.4%	12.3 dB
64	1.12×	1.001×	0.8%	10.4%	10.3 dB

At τ = 1 s the single-channel answer is several times the clock it was meant to measure, and the cross estimate is within a percent of it. The two converge at long τ, and that is correct rather than disappointing: by then the instrument's white PM has averaged down and there is nothing left to reject, which is exactly why the short-τ end is the only place the method is being tested.
Three prices, printed rather than hidden. The estimate is a mean of products and is *signed*: at long τ it goes negative, which means the shared part has fallen below the scatter, and nothing clips it or reports √|var| — a log axis would otherwise grow a noise floor that is not there. Convergence is slow, because only the *estimate* of the instrument noise averages down, as $M^{1/4}$ in amplitude; the rejection column is what was actually achieved, and reduces to the familiar $5 \cdot \log_{10}(M)$ law as a consequence of the error bar rather than as a substitute for one. And correlated channels — a shared reference, a shared supply, one thermal environment — break the method with no symptom whatever in the data, so the independence assumption is a provenance warning on every call.

The error bar cannot be the χ^2 one, a sum of products being unbounded below. Isserlis' theorem gives $(\text{auto}_a \cdot \text{auto}_b + \text{cross}^2)/\eta$, which is $2\sigma^4/\eta$ when the two channels are the same channel — so `cross_variance(a, a)` returns the auto-variance bit for bit and its error bar *is* the chi-square one, rather than a parallel construction that could drift from it.

The two right-hand columns are deliberately not the same quantity, and the gap between them at long τ is the interesting part. "Ensemble spread" is the scatter of the ratio against *this realization's own clock record*, which is what a user comparing two channels of one oscillator actually wants to know; "reported ±" is the standard error about the population variance, which additionally carries the clock's own sampling variation. At short τ the instrument dominates and the two coincide. At long τ the clock dominates, both channels see the same clock record, and the conditional error collapses while the unconditional one does not — so the reported bar is conservative for that question, correctly, since nothing in one measurement can tell it which question was being asked. Against the quantity it does claim — the scatter of the estimate about the truth — it is measured at ±20% out to m = N/4 in `tests/test_cross.py`.

Two estimators were rejected on measurement before this one, and they fail in different places. A Bartlett sum over the products' own autocovariance, assuming no distribution at all, decays monotonically to 0.30 of the true scatter at long τ — the products are heavy-tailed where the coefficients are Gaussian, so each extra lag of their sample autocovariance adds more noise than signal. The equal-shape closed form, which collapses the three correlation functions onto one shape and is what the library shipped first, is fine at both ends of the τ range and 15% narrow in the middle: at short τ the instrument sets all three shapes and at long τ the clock does, so the approximation is exact in both limits and wrong exactly where they mix. The form now used sums all four lagged covariances and approximates nothing; it costs no new numerics, since the cross-correlation of two coefficient arrays comes out of the FFT the EDOF machinery already runs. All three are kept in the test suite and measured on the same realizations, so the reason for the choice stays a live number.

More channels reject deeper, and by less than counting the pairs would suggest. Each of the P = M(M−1)/2 pairs is an unbiased estimate of the same clock, so averaging them helps — but (c0,c1) and (c0,c2) share a channel, so they are not independent estimates and the naive `5·log₁₀(P)`

is an upper bound rather than the answer. Measured at generation time over 40 realizations, as the gain over the two-channel measurement on the same records:

τ (s)	2 channels	3 channels	naive	4 channels	naive
1	18.2 dB	+1.95 dB	+2.39	+3.06 dB	+3.89
2	18.1 dB	+1.60 dB	+2.39	+2.44 dB	+3.89
4	17.8 dB	+0.95 dB	+2.39	+1.39 dB	+3.89
8	16.5 dB	+0.32 dB	+2.39	+0.46 dB	+3.89
16	14.4 dB	+0.08 dB	+2.39	+0.11 dB	+3.89
32	12.3 dB	+0.02 dB	+2.39	+0.03 dB	+3.89
64	10.3 dB	+0.00 dB	+2.39	+0.00 dB	+3.89

> The gain is real at short τ and vanishes at long τ, and both halves are correct. Where the instruments dominate there is noise to reject and the extra pairs reject it; by the time the clock dominates there is nothing left to remove, and a fourth channel buys nothing whatever. A tool that reported the naive figure would promise a constant 3.9 dB across the whole curve and be wrong at both ends of it — overselling the hardware at long τ, and overstating the confidence at short τ, which is the more expensive mistake.

Fitting a model, and what the correlation between τ is worth

Two properties make this more than calling a curve fitter. The sampling law of a stability estimate is **χ^2, not Gaussian**, and least Gaussian at long τ where the specification lives; and the points of an overlapping estimator are **correlated**, because they are computed from the same phase samples. The first is handled by minimizing the exact likelihood — which turns out to be a sum of squares anyway, so an ordinary least-squares optimizer reaches it with no Gaussian approximation. The second is handled by measuring the covariance *between* τ from the coefficient arrays every estimator already returns, and carrying it through a sandwich estimator.

Ignoring the correlation leaves the parameters roughly unbiased and their **uncertainties too narrow** — the one failure mode that survives review, because the fitted value and the residuals both look fine. Measured at generation time over 200 realizations of white FM at a known h_0, N = 4096, where the truth of both fitted parameters is known exactly:

parameter	truth	coverage, correlated	coverage, ignoring it	error-bar ratio
A	1e-22	0.675	0.560	1.26×
b	-1	0.680	0.485	1.37×

> **Nominal coverage is 0.6827 and the Monte-Carlo standard error is 0.033.** The correlation-aware interval sits on nominal; the interval that treats the τ as independent — which is what every other tool reports — misses it by several standard errors, in the direction that overstates confidence. The error-bar ratio is that shortfall seen the other way round.
> The deviance is the other half of the same calibration, and it is reported because an objective whose degrees of freedom are wrong produces intervals that are wrong in the same direction. Measured mean 4.92 ± 0.22 against a predicted 5.02, on 4.7 effective degrees of freedom rather than the 7 a parameter count would give — 9 τ, two of them fitted, and the rest spent on the correlation. Getting that correction wrong is not academic: on a longer record during development, the obvious version — which assumes the fit absorbs p dimensions — predicted 8.0 where the measurement was 5.8 ± 0.4, five standard errors out and in the direction that makes a correct fit look suspiciously good. What the fit actually absorbs is tr(HP), and with correlated points the model lies along the directions where P carries its mass. 0 of 200 fits were flagged as poor by the goodness-of-fit test; all of them are still scored above, since dropping them would be selecting on the outcome being measured.
>
> Neither number is available from a parameter count, and neither is available at all without the cross-τ covariance — which no other tool in this field computes.

Design decisions worth recording

Scale factors live outside the coefficients

Every estimator returns the coefficient array it averaged, so one confidence-interval implementation can serve all of them. The obvious form is `variance = mean(c²)`, which requires dividing by √2 or √6. That was implemented and then reverted: it makes every coefficient irrational and destroys exactness where the mathematics is exact — the drift identity above failed by one bit. The contract is now `variance = scale · mean(c²)`. EDOF depends only on the scale-free correlation structure, so nothing is lost.

float64 phase is a genuine hazard at this precision

At an accumulated phase of 10^3–10^6 s the float64 spacing is 10^{-13}–10^{-10} s, which silently quantizes away exactly the picoseconds sub-nanosecond work cares about. Phase is therefore stored as offsets from an explicit epoch, and construction warns when the spacing approaches the smallest resolved step. The warning is recorded in provenance, not merely printed, so it survives into a report.

The classical estimators are not routed through the wavelet filter bank

The Haar-equals-Allan identity is the intellectual core of the library, but making it the *implementation* would put boundary handling and shift-equivariance on the critical path for no user-visible gain. Written directly, the identity becomes an independent check of two separate code paths instead of a tautology.

No pickle support, by design

`pickle.load` on a file arriving from a customer's test rig is arbitrary code execution. `.npy` covers the same need safely.

Current state

test suite	`[32m[32m[1m1297 passed[0m[32m in 296.30s (0:04:56)[0m[0m[0m PASS`
wall time	`323.2 s`
latest commit	`2969c94 — Find discrete tones, measure them exactly, and remove them loudly`

Three structural properties are enforced by tests rather than convention: the core computes with scipy and matplotlib absent; every theory page must declare its validation status and name a covering test; and built-in readers and estimators register through the same decorator a third-party plugin would use.

Built

- **M0** — immutable phase/frequency containers with the precision guard, provenance with a step log, the plugin registry, tau selection, Kasdin–Walter synthesis, delimited-text reading, theory pages, and a CLI.
- **M1** — `adev`, `oadev`, `mdev`, `tdev`, `ohdev` computed directly from phase, gap-aware, plus the exact-rational oracle and the Tier 0 identity suite.
- **M2** — η_1 and η_3 degrees of freedom estimated from the estimator's own coefficient autocovariance, the pure-numpy incomplete gamma and inverse χ^2, χ^2 confidence intervals, lag-1 noise identification with an MDEV-slope cross-check, the deviation bias correction, and `stabilis validate` emitting a versioned evidence report that is capable of failing. This is the v0.1 line: before it, no uncertainty could be quoted from the library at all.
- **M3, first piece** — Greenhall EDOF, transcribed from the Greenhall & Riley PTTI 2003 preprint: all four cases, three coefficient tables and the generalized autocovariance, plus Satterthwaite pooling over the runs of a gapped record. It is what Stable32's `Alpha`, `Min Sigma` and `Max Sigma` columns are computed from, so it is what makes an old report and a new one diffable line for line. It is selected explicitly and never by `auto`: assuming a power law has to be a recorded act, and where the noise cannot be identified from the data the library declines and falls back rather than guessing, which is a place it deliberately answers less than Stable32 does.
- **M3, second piece** — the **power-law decomposition**, which reads a measured $\sigma(\tau)$ curve back into the five IEEE 1139 noise levels and a drift term. Because every h_a is a noise power and drift enters as D^2, the model is linear with a non-negativity constraint, which is exactly non-negative least squares — solved exactly by Lawson–Hanson, with a KKT optimality certificate on every fit, rather than by an optimizer that could be blamed for a negative answer. It fits Allan and *modified* Allan rows together, because white PM and flicker PM differ under AVAR only by a slowly varying logarithm: on noiseless data that is enough, but with 2% noise on the curve an AVAR-only fit returned between 0.00 and 1.00 times the true h_1 across six seeds, where the stacked fit held to 5% on every one. The conversion coefficients behind it were transcribed from NIST SP 1065 §7.1, whose flicker-PM row is printed two mutually inconsistent ways — resolved by integrating the handbook's own transfer function, and recorded rather than silently corrected. The five modified-Allan coefficients the handbook does not tabulate at all were derived here from its eq. (66).
- **Amplitude-calibrated synthesis** — the noise generator's h_α is now a settable parameter in SI units rather than an arbitrary scale, checked against four closed forms that are exact at every

averaging factor. That is what makes the decomposition testable end to end: noise is generated at a known level, estimated, decomposed, and compared to what went in.

- **M3, the estimator tail** — **TOTVAR** with its published EDOF and the SP 1065 bias correction; the **Thêo family** — Thêo1, ThêoBR and ThêoH, reaching 75% of the record where the Allan variance stops at half; **MTIE and TIE rms** by binary doubling, 10^6 points over 18 octaves in 0.15 s; and the **MODWT wavelet variance**.
- **The wavelet layer**, which is the one that reframes the rest: the Haar wavelet variance *is* the Allan variance, exactly rather than approximately, so the classical estimator is the K=1 member of one family and the Daubechies filters are the remainder. The library asserts that identity to 1e-14 over eleven octaves and again on the confidence interval, where the Haar bounds come back equal to the overlapping Allan bounds. What the higher orders buy is concrete: K vanishing moments annihilate any polynomial trend of degree below K, so D4 removes a linear frequency drift with no detrending step at all — otherwise a separate and contestable operation on a customer's data. The filter coefficients are computed by spectral factorization and then certified against the defining algebra, rather than transcribed, because a transcribed list with one wrong digit still looks like a filter.
- **Three misprints in NIST SP 1065 found by measurement**, all recorded rather than silently corrected: the flicker-PM conversion row printed two inconsistent ways; the 0.75 missing from the Thêo1 equation; and Thêo1's averaging time, printed as $0.75(m-1)\tau_0$ once and $0.75m\tau_0$ twice, where the handbook's own ThêoBR equation only parses under the second and measurement puts the first 10% low at m=10. For a tool whose product is correctness, being able to show the evidence for each departure is the deliverable, not a footnote.
- **The output and interaction layer** — plotting, a flat 30-name public API, and `stabilis gui`, a local browser page in four stages that computes nothing itself: every variance and bound on it comes from the same calls the CLI makes, and every selection prints the equivalent non-interactive script, because an exclusion that exists only as a mouse gesture is not evidence. The fourth stage is the bench, which is the only one that reads several files at once and therefore the one that assumes least about them.
- **M5 complete, ahead of its place in the sequence.** The **bench model** describes the actual experiment as sources, devices and channels, and turns three-cornered hat, N-cornered hat and instrument noise-floor subtraction into one non-negative linear solve per τ rather than three features. The part with no equivalent anywhere is the refusal: identifiability is a graph property — a connected set of measurements determines its oscillators individually *iff it contains an odd cycle*, which is why the technique is called *three*-cornered hat — so a ring of four determines nothing however long it runs, and `stabilis bench check` says so *before any data is collected* and names the one extra measurement that would fix it. It also distinguishes the pairs you must go and measure from the pairs you already have, since $x_{ab} = x_{ac} - x_{bc}$ identically: two clocks against a common reference are one subtraction away from the edge that closes the triangle, and a four-clock ring needs no new data at all.
  The **cross-variance** is the other half of the same idea, one level down. Two channels watching one clock: the auto-variance of either is floored at $\sigma^2_c + \sigma^2_n$ however long you log, but the difference operator is linear, so $E[d_a d_b] = E[d_c^2]$ when the channels' noise is independent — the instrument term has expectation zero rather than being small. It costs one function for the whole estimator family, because every result already carries the coefficient array it averaged and the

cross case is `mean(c_a·c_b)` where the auto case is `mean(c²)`. Its error bar cannot be the χ^2 one, a sum of products being unbounded below; Isserlis' theorem gives (auto_a·auto_b + cross²)/η instead, which is $2\sigma^4/\eta$ when the two channels are the same channel — so `cross_variance(a, a)` returns the auto-variance bit for bit and the two error bars are one construction rather than two. `cross_matrix` does every pair at once and `reject` averages them, which reaches deeper still; the pairs share channels and are therefore not independent, and the exact covariance says how much of the naive gain is real. Negative variances are reported signed rather than clipped throughout, because they are the method telling you an independence assumption failed.

- **The measurement report**, which is the actual paid deliverable and now exists: one self-contained HTML document per measurement, carrying the σ(τ) table with its intervals, the noise type at each τ, the record's own hash, the full provenance step log, and a conformity verdict. The verdict is the part that differs from every other tool — **it is taken on the confidence interval, not on the point estimate.** A point passes only when its whole interval lies below the limit and fails only when the whole interval lies above it; an interval straddling the limit is reported *indeterminate*, meaning the record is too short to decide and the remedy is more data. Comparing the point estimate instead — what the field normally does — is simple acceptance with a ~50% false-accept risk for a unit sitting on the limit, and it is worse than that here, because at low degrees of freedom the point estimate is itself biased low (11.4% at η = 2) and the error therefore runs towards passing exactly where the evidence is thinnest. Three rules follow and each is a test: a limit is never interpolated between the taus a datasheet states, no interval means no verdict, and a two-source pass is conclusive while a failure is not attributable.
- **Model fitting under the χ^2 likelihood**, with the correlation between τ measured rather than assumed away. Arbitrary models — a Gauss–Markov correlation time, a servo bump, a quantization floor, or a customer's own — fitted against the exact sampling law rather than by least squares, which is the wrong objective for a distribution running from 0.55σ to 2.2σ. Two independently derived uncertainties come back: a correlation-aware sandwich and a copula bootstrap that keeps both the χ^2 margins and the measured correlation. They agree when the model is right and disagree when it is not, which is why both are printed. And the fit **refuses to quote parameters** whose deviance is outside what its effective degrees of freedom admit — in either direction, since a fit that is too good means the model has more freedom than the curve constrains.
- **The export layer** — the last thing between the library and being usable inside somebody else's workflow, since until now nothing in the package wrote data at all. A written record **carries its own rate and epoch**, so a round trip needs no arguments and cannot be given the wrong ones; it is written at 17 significant digits, because `%.15e` returns a different double for 42% of them, measured. The Stable32 block is transcribed rather than derived and then checked against Stable32's own bytes: all 6461 committed data rows are reproduced character for character, the eight preamble lines with them, and the σ column computed here from the same input agrees to the last digit the format prints. Four refusals rather than inventions — an estimator Stable32 never named, a frequency series whose preamble has never been captured, a gapped record where the preamble asserts contiguity, and a result with no α, since writing 0 there would assert white FM. And `stabilis export bundle`: a directory holding the data copied byte for byte, the curve, the report, the provenance and a SHA-256 for each, whose **digest is over the measurement rather than the files** — hashing the files would have been the obvious choice and is wrong,

because the provenance embedded in them carries a timestamp and a digest that changes every run says nothing.

- **Power spectra** — Welch, one-sided, with the window's noise-equivalent bandwidth divided out so the result is a density rather than a periodogram in disguise; checked by Parseval to twelve digits and by recovering a synthesized h_0 at three amplitudes over six decades. Two conversions are *refused* rather than guessed: S_φ without a carrier frequency, since a 10 MHz OCXO and a 10 GHz source give identical time-error records at equal fractional stability; and $\mathscr{L}(f) = S_\varphi/2$ wherever the integrated phase excursion is too large for the small-angle limit, which is precisely the near-carrier region a datasheet is written from. Two findings came out of the measurements rather than the reasoning: the mean is the unbiased statistic and the median sits 3.4% low at 15 segments, and a cumulative sum is not an ideal integrator — the two differ by $(\pi f \tau_0)^2/\sin^2(\pi f \tau_0)$, rising to 2.467 at Nyquist.
- **Limit masks with a verdict** — ITU-T G.811 transcribed from the Recommendation itself, which changed a number: the widely repeated form of its MTIE limit above 1000 s is a flat 300 ns ceiling, and the document says $10^{-5}\tau$ + 0.29 μs, which rises. The *measurement conditions* the standard states are enforced as part of the limit, so a record sampled at 1 Hz or shorter than the 12τ it demands returns "not evaluated" rather than a green tick. An extreme-value statistic is judged differently from a variance: exceeding an MTIE limit fails conclusively without an interval, while staying under one is a pass over the measured window only, since a longer record can only find a larger maximum.
- **Spurs** — found against a running-median floor, then *measured by fitting the record* rather than reading the spectrum, so a tone between bins is exact without a flat-top window: 0.001% on amplitude and about a nanohertz on frequency. Every table prints its own expected false-alarm count from the chi-square tail. Notching records each removed tone in the returned series' provenance, because a record with a tone taken out describes an oscillator that does not exist.

Next

- **The rest of the Stable32 session** — real 1.62 exports over six synthetic records are committed and scored automatically, so the claim is no longer "implements the published algorithm" but "matches Stable32 to the precision it prints". Still to come: the public NBS suites and, most valuable of all, a real oscillator record, being the only case where the data was not generated by the tool under test.
- **Somebody else opening a file we wrote** — the export layer's round trips all go through this library's own reader, which shows the pair is self-consistent and nothing more. The mirror of the rule already in force for readers: a writer validated only against its own reader is validated against nothing. It closes with one line added to the Stable32 run sheet, and is recorded as item 8 of `VALIDATION-DEBT.md` until it does.
- **M4–M8** — the rest of M4 is instrument readers, gated on paired vendor exports rather than on effort: a binary format with no public specification can only be accepted if the same capture is also available as vendor ASCII to check the reader against. The rest of M8 is the interactive span selector and the remaining plots. Then realtime ingest with attributed time tagging, and dynamic Allan variance.

> **Known limitations.** Confidence intervals now exist, but their measured coverage falls short of nominal at the extremes of the τ range; the measured figures live in `docs/validation-report.html`, generated by `stabilis validate`, which is the artifact that carries them. Greenhall's bounds are now available alongside the data-driven ones, but have now been checked against Stable32 1.62's own exports on this project's six synthetic records: the term-count column agrees exactly as integers, the sigma columns of all five estimators agree to 4.9e-5 — the last digit Stable32 prints — and Min/Max Sigma agree to 5.5e-4. The degrees of freedom behind those bounds do *not* quite agree: Stable32's implied EDOF runs about 1.2% above ours for averaging factors up to 32, where an independent implementation of the same paper returns our value to every printed digit. Cause unknown, worth under 0.06% on a 68.27% bound, and recorded rather than guessed at. No real oscillator record has been through both programs yet, which is the remaining part of that item in `VALIDATION-DEBT.md`. Of the ten power-law conversion coefficients, nine are checked end to end against noise of known amplitude; the tenth, unmodified flicker PM, cannot be with the current generator, because that coefficient assumes a brick-wall bandwidth and the Kasdin–Walter process is not brick-wall limited — the gap is measured, bounded and recorded rather than averaged away. Instrument readers are not yet validated against real hardware exports, and are deliberately not written until a paired vendor export exists to check them against. The Thêo family has no confidence interval: it reports no coefficient arrays for the data-driven path and its own EDF table is not implemented, so the CLI offers no interval flags for it rather than flags that would fail. ThêoBR's bias correction is measured from the data and is itself noisy on a single record — a common vertical shift of the whole curve, not scatter between points — so the ratio is reported in provenance rather than folded silently into the answer. The conformity verdict inherits its interval's accuracy: the decision rule is fully tested at both boundaries and the intervals have measured containment, but containment is an ensemble statement while a verdict is issued on one record, so the two do not compose into a stated false-accept risk — what is missing is an operating-characteristic curve, and until it exists the report states the rule and the interval it used rather than a risk figure. The cross-variance has no interop tier and can never have one, since no other tool computes it and manufacturing a reference from this library's own output would make the assertion a statement about itself; its evidence is correspondingly heavier on the functional side, and the hardware half — two real channels, deliberately sharing a reference in one run and not in another — is still owed.